

WILLIAM GEMBA

AI/ML Engineer

Miami - Fort Lauderdale Area | Cell: 917.588.1702 | Email: gemba.william@gmail.com | Social: [linkedin.com/in/william-gemba](https://www.linkedin.com/in/william-gemba)
Portfolio: wgemba.github.io/webportfolio/ | Medium: medium.com/@gemba.william

SUMMARY

AI/ML Engineer with depth in production machine learning and applied AI for financial services. Proven track record designing and deploying end-to-end ML solutions on AWS (including time-series forecasting with full MLOps lifecycle, RAG knowledge systems, and multi-agent platforms) that automate complex analytical workflows. Combines ML engineering and data science depth with hands-on backend development and domain expertise in banking and finance.

TECHNICAL SKILLS

Programming & Frameworks: Python, SQL (Postgres, Oracle), JavaScript, Node.js, HTML/CSS, Flask, FastAPI
AI/ML & Data Science: Retrieval-Augmented Generation (RAG), Agentic AI, Agent Orchestration (Strands Agents SDK), Intelligent Document Processing (IDP), LLM Integration (Anthropic Claude, Nova, Amazon Bedrock), Prompt Engineering (Few-Shot, CoT, etc.), Time Series Forecasting (ARIMA, Prophet, LSTM, DeepAR), XGBoost, Scikit-Learn, LightGBM, SHAP, Pandas, NumPy
Cloud Platform & Infrastructure: AWS (Bedrock, SageMaker, AgentCore, Lambda, S3, EC2, ECR, Glue, Athena, Redshift, Step Functions, DynamoDB, Kendra, Bedrock Knowledge Bases, CloudWatch, X-Ray, OpenTelemetry)
MLOps & Engineering: Docker, GitHub Actions (CI/CD), MLflow, SageMaker Experiments, Streamlit, ETL Pipeline Design, Feature Engineering, Model Evaluation & Monitoring, Terraform (IaC)

RELEVANT WORK EXPERIENCE

AI/ML Engineer | BankUnited | Miami, FL May 2024 - Present

- Architected and deployed a multi-agent credit analysis platform (Strands Agents SDK) using a hub-and-spoke pattern with specialized financial, risk, and industry analysis agents, reducing credit analysis cycle time and accelerating Credit Approval Memorandum drafting.
- Developed a deposit forecasting tool for the Commercial Banking unit using a point-in-time LightGBM approach with engineered temporal features, validated via a 12-month backtest; deployed on SageMaker with Feature Store, Model Registry, real-time endpoint, and drift-triggered automated retraining via Model Monitor.
- Engineered an intelligent document processing pipeline that classifies and extracts structured data from financial documents (SEC filings, audited reports, pro-forma statements, CRE statements, etc.) via schema-based extraction mapped to a classification-driven registry.
- Integrated internal credit policy knowledge bases and third-party industry data APIs (IBISWorld) into the agentic workflow, enabling automated credit risk narrative generation benchmarked against bank policy and peer group performance.
- Developed an enterprise RAG knowledge system for client services using Claude LLMs with Bedrock Knowledge Bases (Kendra GenAI indices), implementing embedding strategies, retrieval optimization, and prompt engineering via few-shot learning and Chain-of-Thought reasoning, reducing manual lookup workloads by 42%.
- Delivered full-stack AI applications with Python backends and JavaScript/Node.js frontends, containerized with Docker, deployed on EC2 via ECR, served behind an application load balancer, and automated through GitHub Actions CI/CD pipelines.

Technology Consultant | Ernst & Young (EY) | New York, NY June 2021 - June 2023

- Delivered ETL development, SQL optimization, and data lineage documentation for Tier 1 financial services clients, supporting regulatory reporting modernization and audit readiness under federal regulatory body specifications.

PROJECTS

Mortgage Credit Default Intelligence System | Python, XGBoost, SHAP, SageMaker, FastAPI, Strands Agents SDK, Bedrock

- Built an end-to-end mortgage default prediction system on 1.86B rows of Fannie Mae loan-level data (2010-2025), engineering 28 borrower, collateral, performance, and macro-regime features via Athena over S3 Parquet, and benchmarking logistic regression, XGBoost, and LightGBM with MLflow tracking and Model Registry stage promotion.
- Designed a regime-change stress test framing the 2022+ rate shock as out-of-time evaluation; champion XGBoost model achieved 0.909 AUC with 4.4-point degradation post-2022, and Population Stability Index analysis isolated macro-rate features (PSI 5.8-11.8) from stable credit-behavior features (PSI < 0.1), producing a two-tier production monitoring framework.
- Deployed the champion model to a SageMaker real-time endpoint behind a FastAPI service with per-prediction SHAP explanations, and built an agentic credit analyst layer (Strands SDK, Claude on Bedrock) that retrieves Fannie Mae Selling Guide policy via RAG and generates comprehensive credit memos grounded in model outputs and policy citations.

CERTIFICATIONS

AWS Certified Machine Learning Engineer – Associate, Amazon Web Services Issued June 2025
Machine Learning Specialization, DeepLearning.AI & Stanford University Issued December 2023

EDUCATION

Fordham University
MS in Data Science | Advanced Certificate in Financial Econometrics & Data Analysis
BS in Finance & Business Economics